

Subject Description Form

Subject Code	SFT5108
Subject Title	Green Innovations for Fashion and Textiles
Credit Value	3 credits
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	Textiles (clothing, home, and technical variants) universally impact every aspect of our modern world. Unfortunately, across the textile life cycle, there are many potential detrimental impacts on the environment as a result of the various syntheses, waste outputs etc. Due to the combined drivers of climate change fears, interests in sustainability, increased government legislation as well as general desires for improved quality of life, every aspect of the textile life cycle, from fabrication, to sale, repair as well as end-of-life waste repurposing, is currently the focus for improvement and innovation, worldwide. This course aims to introduce and help students better understand these new innovative approaches, often grounded in novel applications of science and engineering principles, for use on textiles and related soft matter. Broadly we will cover the applied science process, from the utilisation of known materials properties to best meet a given sustainability purpose. This will be done through introduction of a series of ‘hot topics’ or themes, in order to explore the fundamental principles as well as the various real-world applications across a multitude of disciplines. From progress on the course, students will begin to obtain a fuller understanding of the drivers for certain changes in the textiles field, including how important new scientific developments are facilitating changes in textiles fiber and fabric engineering, pre-treatment, colouration, finishing, retail, waste treatment, waste minimisation, and recyclability. In addition, new treatment processes and fabrication methods are facilitating applications in industrial, design, conservation and archi-textiles fields, for enhanced performance, energy-efficiency and/or improved outputs, will be explored. As a result, students will gain an appreciation from a broad overview of many of the important textile technology and soft matter materials science touchstones of our time and lay the groundwork for planning applications for various green innovations in order to best meet social, environmental, engineering and other overlapping needs. Throughout the course students are expected to read a range of English-language references and case studies in order to build and consolidate their knowledge for comprehending and being able to apply contemporary findings and new directions in sustainable textile technology. The course encourages the creation of new “out of the box” ideas, in order to enhance critical, creative, and innovative thinking. An extended piece of writing and group coursework projects will allow expression of such ideas in the course of fulfilling a specified technical project brief.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the characteristics of various technologies/innovations/advances in engineering, conservation, materials science, etc, that define eventual potential applications across a multitude of

	fashion and textiles applications (b) understand a range of principles in science and engineering in order to make well-founded decisions as part of a design process for green innovation in new textile engineering development applications (c) understand the use of appropriate candidate technologies in the planning and utilisation steps towards a desired output, in accordance with a given target brief specification (d) develop critical and creative skills in analysing, explaining and ideally matching innovations in sustainability to specific textile technology applications, based on insights and knowledge gained from lectures, readings and own researches into the subject
Subject Synopsis/ Indicative Syllabus	This subject will be delivered predominantly via interactive lectures, complemented by sessions held at the HK PolyU Materials Resource Centre (housed in the HK PolyU School of Design). 1. Introduction to green innovations in soft matter & textiles + Course overview + Assignment formats & details 2. Chemicals, colourants and the environment¹⁻⁵ ▪ Use of biotechnology (e.g.; algae, bacteria, enzymes, etc.) as new alternatives to conventional colouration methods 3. Fabrics and fashion⁶⁻⁹ ▪ Developments in new synthetics (e.g.; leathers) and advances in 3D printing 4. New technologies and directions in textile conservation^{10,11} ▪ Self-healing & shape-changing polymers and gels – potential applications in, e.g.; textile art conservation 5. Materials applications in design and architecture¹²⁻¹⁴ ▪ Combining conventional textile manipulation and manufacturing methods with smart materials (intelligent reactive) for improved strength/durability/performance in novel archi-textiles for enhanced energy efficiency 6. Catalytic degradation by and of textile waste materials^{15,16} ▪ Exploring both the use of clothing as a means of degrading pollutants, as well as related mechanisms that can degrade textile processing waste 7. Environmentally-responsive textiles for thermal comfort¹⁷⁻²⁰ ▪ Improving wearer thermal comfort and reducing energy usage on environmental heating and cooling, through the use of thermo-responsive clothing and textiles 8. Recycling and sustainability in the textiles industry²¹⁻²⁷ ▪ New methods of re-using, re-purposing and re-generating waste by-products, both from the textile industry and beyond, in order to make new materials and reduce textile waste outputs from the industry. 9. Wearables and VR for green innovations²⁸⁻³⁰ ▪ Advancements in VR, AI and AR, as facilitated by haptics-based wearables, will have profound impacts on all aspects of the textile and clothing industry. We will explore such impacts in this topic as well as the method by which such technology is facilitated.
Teaching/	• Lectures will use various visual and written references to facilitate

Learning Methodology	student's understanding of the subject • A summary of each lecture's content will be distributed to students on a weekly basis • In order to engage student's participation, they are required to prepare pre-class material that is ready for subsequent discussion (~3 mins) related to the topics to be covered during the lecture – random students/groups will be chosen to present to the whole class. In addition, a short MC test will also need to be completed after the lecture, to reinforce that week's topic. • Substantial mandatory reading assignments will be given weekly, relating to the lecture topic of contemporary interest																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1" data-bbox="395 562 1444 976"> <tr> <th data-bbox="395 562 815 674" rowspan="2">Specific assessment methods/tasks</th><th data-bbox="820 562 967 674" rowspan="2">% weightin g</th><th colspan="4" data-bbox="971 562 1444 674">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th data-bbox="971 680 1086 707">a</th><th data-bbox="1091 680 1206 707">b</th><th data-bbox="1211 680 1326 707">c</th><th data-bbox="1331 680 1444 707">d</th></tr> <tr> <td data-bbox="395 714 815 748">1. Group Project Assignment</td><td data-bbox="820 714 967 748">20%</td><td data-bbox="971 714 1086 748">✓</td><td data-bbox="1091 714 1206 748">✓</td><td data-bbox="1211 714 1326 748">✓</td><td data-bbox="1331 714 1444 748">✓</td></tr> <tr> <td data-bbox="395 754 815 822">2. Individual Project Assignment</td><td data-bbox="820 754 967 822">20%</td><td data-bbox="971 754 1086 822">✓</td><td data-bbox="1091 754 1206 822">✓</td><td data-bbox="1211 754 1326 822">✓</td><td data-bbox="1331 754 1444 822">✓</td></tr> <tr> <td data-bbox="395 828 815 896">3. Pre-class preparation, attendance & participation</td><td data-bbox="820 828 967 896">20%</td><td data-bbox="971 828 1086 896">✓</td><td data-bbox="1091 828 1206 896">✓</td><td data-bbox="1211 828 1326 896">✓</td><td data-bbox="1331 828 1444 896">✓</td></tr> <tr> <td data-bbox="395 902 815 936">4. Exam</td><td data-bbox="820 902 967 936">40%</td><td data-bbox="971 902 1086 936">✓</td><td data-bbox="1091 902 1206 936">✓</td><td data-bbox="1211 902 1326 936">✓</td><td data-bbox="1331 902 1444 936">✓</td></tr> <tr> <td data-bbox="395 943 815 976">Total</td><td data-bbox="820 943 967 976">100 %</td><td colspan="4" data-bbox="971 943 1444 976"></td></tr> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: GROUP PROJECT ASSIGNMENT & IN-CLASS PRESENTATION: 20% In small teams, students will build a technical poster regarding a chosen topic in green innovation technology. Subsequently, students will also provide a short oral presentation overview (including Q&A section) of their chosen topic. This will help develop and improve the student's level of literacy as well as higher-order critical and creating thinking. This will help develop and improve the student's level of higher-order critical and creating thinking, abstraction and presentation abilities. INDIVIDUAL WRITING ASSIGNMENTS & IN-CLASS PRESENTATION: 20% Students must perform develop an extended piece of writing (~2500 words) regarding a chosen topic in green innovation technology. Subsequently, students will also provide a short oral presentation overview of their chosen topic. This will help develop and improve the student's level of literacy as well as higher-order critical and creating thinking, advanced literacy and comprehension and abstraction abilities. ATTENDANCE, PRE-CLASS PREPARATION & PARTICIPATION, PRESENTATIONS: 20% Pre-class activities (e.g.; MC tests & short discussion topics questions preparation) [10% each]; class attendance [5%], & active participation during the lecture presentation will be assessed (e.g.; via '<i>clickers</i>' etc.) [5%]. EXAM PAPER: 40%					Specific assessment methods/tasks	% weightin g	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Group Project Assignment	20%	✓	✓	✓	✓	2. Individual Project Assignment	20%	✓	✓	✓	✓	3. Pre-class preparation, attendance & participation	20%	✓	✓	✓	✓	4. Exam	40%	✓	✓	✓	✓	Total	100 %				
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	A 3 hour, end-of-term exam paper comprising multiple long answer questions will test breadth of knowledge and comprehension across the entirety of the course.		
Student Study Effort Expected	CLASS CONTACT:		
	▪ Lectures	25 hrs. (i.e.; 10*2.5 hrs)	
	▪ Tutorials	14 hrs. (i.e.; 4*3.5 hrs)	
	OTHER STUDY EFFORT:		
	▪ Individual Assignment (<i>Due Week 6</i>)	24 hrs.	
	▪ Group Assignment (<i>Due week 12</i>)	22 hrs.	
	▪ Self-study & Class Preparation	20 hrs.	
	TOTAL STUDENT STUDY EFFORT	105 hrs.	
Reading List and References	1. Holkar, C. R., Jadhav, A. J., Pinjari, D. V., Mahamuni, N. M. & Pandit, A. B. A critical review on textile wastewater treatments: Possible approaches. <i>J. Environ. Manage.</i> 182 , 351–366 (2016).		
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- Properties and Applications* 442–71 (Woodhead Publishing, 2015). doi:10.1016/B978-1-78242-380-5.00016-9
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